Abhishek Raj

HPC Application Engineer

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Summary

High Performance Computing (HPC) Engineer with 2+ years of experience designing, deploying, and supporting Linux-based supercomputing infrastructure under India's National Supercomputing Mission. Experienced in HPC platform engineering, scientific software deployment, parallel programming (MPI/OpenMP/CUDA), Linux administration, and containerized research environments. Passionate about building scalable computing platforms for research and AI workloads.

Skills

Programming and Parallel Computing: C, C++, Python, Bash, Fortran, MPI, OpenMP, CUDA, OpenACC
Linux & System Administration: Linux (CentOS, Ubuntu, Arch Linux), Shell Scripting, User Management, QoS, LFS, SSH, System Troubleshooting
HPC Platform Engineering: SLURM, Spack, Conda, Modules, JupyterHub, Docker, Apptainer (Singularity), HPC Software Deployment, Cluster Monitoring
HPC Toolchains & Applications: GCC, Intel oneAPI, Intel MPI, OpenMPI, NVHPC, CUDA Toolkit, MKL, FFTW, HDF5, NetCDF, LibXC, VTune, Gaussian, Quantum ESPRESSO, GROMACS, LAMMPS, WRF, VASP
Infrastructure & Research Computing: Application Porting, Software Debugging, Dependency Resolution, Performance Analysis, AI/ML Environment Deployment, Technical Documentation, Git
Familiar With: Kubernetes, Ansible, Jenkins, Prometheus, Grafana, Batch Scripting, Perl Scripting, JAVA

Experience

Project Engineer (HPC Application Engineer) **Apr 2024 – Present**
Centre for Development of Advanced Computing (C-DAC), Pune *Deputed at IIT (BHU), Varanasi*

- Administer and maintain the **225+ node PARAM Shivay** supercomputing facility under the National Supercomputing Mission (NSM), including **11 NVIDIA Tesla V100 GPU nodes**, supporting AI/ML, computational chemistry, materials science, CFD, and engineering workloads.
- Build, deploy, optimize and maintain **100+** HPC software environments using **Spack** and **Conda**, maintaining applications including **GROMACS, Gaussian, Quantum ESPRESSO, VASP, MPI, CUDA, Python**, and AI/ML frameworks.
- Engineer Linux-based HPC environments by resolving compiler, MPI, library, and software dependency issues, optimizing application performance, and preparing reproducible research environments.
- Develop technical documentation, installation guides, and cluster usage policies while providing HPC platform support and troubleshooting for researchers utilizing India's **PARAM** supercomputing facilities.
- Delivered **Parallel Programming** training for **two successful HPC Application Programming batches** at **C-DAC ACTS**, covering MPI, OpenMP, CUDA, and High Performance Computing through lectures and hands-on laboratory sessions.
- Conduct workshops and mentor **500+** students, researchers, and HPC users at **C-DAC** and **IIT (BHU)** on Linux, parallel programming, and scientific computing technologies.

Projects

JupyterHub Deployment & HPC Training Platform | *Linux, JupyterHub, HPC* **2025**

- Designed and deployed a production-ready **JupyterHub** platform as a **systemd** service on **PARAM Rudra** and other PARAM supercomputers with **LDAP**, **SLURM**, and multiple Jupyter kernels.
- Developed reusable HPC environments featuring **MPI, OpenMP, CUDA, C/C++, Python**, AI/ML frameworks, and scientific applications for research, training, and browser-based HPC access.
- Prepared deployment guides, hands-on laboratory exercises, and technical documentation adopted during HPC training programs at **C-DAC** and **IIT (BHU)**.

HPCCFS: Building an HPC Cluster From Scratch | *Linux, SLURM, VirtualBox* **2026**

- Designed an open-source guide to build a production-style HPC cluster from scratch using **VirtualBox**, covering **SLURM**, **MUNGE**, **NFS**, **OpenLDAP**, **QoS**, networking, and Linux cluster administration.
- Documented cluster provisioning, **Spack**, environment modules, MPI runtime, scheduling policies, and automated user management for reproducible HPC environments.

Education

PG-Diploma in High Performance Computing Application Programming (PG-DAC HPCAP) **2023 – 2024**
C-DAC ACTS, Pune *Grade: A (84.71%)*

Bachelor of Technology (Computer Science & Engineering) **2018 – 2022**
Darbhanga College of Engineering, Darbhanga, Bihar *CGPA: 7.80/10*